

What If the Vacuum Gravitates?

A Reinterpretation of Λ CDM That Might Resolve Its Paradoxes

Boris Kriger^{1,2}

¹ Information Physics Institute, Gosport

boris.kriger@informationphysicsinstitute.net

² Institute of Integrative and Interdisciplinary Research, Toronto

boriskruger@interdisciplinary-institute.org

ORCID: <https://orcid.org/0009-0001-0034-2903>

Abstract

We propose a framework in which the cosmological constant Λ and the quantum vacuum energy density ρ_{vac} are treated as physically distinct quantities. The cosmological constant is returned to its original Einsteinian role as a geometric property of space-time, constant everywhere, driving the uniform expansion of space. Vacuum energy is placed in the stress-energy tensor $T_{\mu\nu}$ as a local form of matter-energy that gravitates, varies with cosmic expansion, and responds to the local matter density through the ansatz $\Lambda(\rho_m) = \Lambda_0 - \alpha \rho_m$, where $\alpha \sim 0.01\text{--}0.05$ is constrained by nucleosynthesis, matter dilution, and running vacuum model data.

Inside gravitationally bound structures, where cosmic expansion is suppressed, vacuum energy gravitates without compensation—producing an invisible, collisionless, uniformly distributed gravitating component whose qualitative properties match those attributed to dark matter. The halo boundary is identified as a phase transition between bound and expanding vacuum. The total gravitational mass of a galaxy consists of three components: visible baryonic matter, invisible compact remnants (whose population remains observationally unconstrained in significant mass ranges), and gravitationally bound vacuum energy. We do not specify the proportions; we assert that their sum reproduces observed rotation curves, gravitational lensing, and cluster dynamics.

In the early universe, when the vacuum energy density was higher by a factor of $(1+z)^3$ (a scaling selected by self-consistency, not assumed as a free parameter), the enhanced local gravitational environment accelerated structure formation, providing a natural mechanism for the rapid assembly of supermassive black holes observed by JWST at $z > 7$. Vacuum energy does not enter the Friedmann equation—it is sequestered from cosmological dynamics while remaining gravitationally active inside bound structures—ensuring compatibility with CMB observations. The filamentary cosmic web is reinterpreted as a dynamic structure at the intermediate scale between individual galaxies and the Hubble volume, stabilized by the interplay of cosmic expansion, peculiar velocities, and vacuum energy gradients.

The model reinterprets the physical content of Λ CDM—not its mathematics—and reframes the cosmological constant problem, the non-detection of dark matter particles, the cosmic web morphology, and the early supermassive black hole puzzle as aspects of a single underlying issue: the 1967 identification of Λ with ρ_{vac} , which was assumed without proof and has produced a 10^{120} -order crisis.

Keywords: vacuum energy, cosmological constant, dark matter, cosmic web, cosmic voids, galactic rotation curves, supermassive black holes, Λ CDM reinterpretation, gravitational structure

Contents

References	4
1 Introduction	5
2 Logical Structure of the Argument	6
3 Background: The Vacuum Has Energy	6
3.1 Structural Evidence from the Standard Model	6
3.2 The Zel'dovich Identification and Its Consequences	7
4 Conceptual Framework: Separating Λ and Vacuum Energy	8
4.1 Reframing the Cosmological Constant Problem	8
5 Mathematical Framework	9
5.1 The $\Lambda(\rho_m)$ Ansatz	9
5.2 Modified Field Equations	10
5.3 Modified Poisson Equation and Effective Gravity	10
5.4 Why Vacuum Energy Does Not Enter the Friedmann Equation	11
5.5 CMB Compatibility	11
5.6 Vacuum Energy in the Early Universe	12
5.7 Self-Consistent Fixed-Point Structure	12
6 Bound Versus Expanding Vacuum	13
6.1 The Fundamental Observation	14
6.2 Expanding Phase (Voids)	14
6.3 Bound Phase (Inside Galaxies)	14
6.4 The Phase Transition Boundary	14
6.5 Order-of-Magnitude Estimate	14
7 Three-Component Model of Galactic Gravity	15
7.1 The Three Components	15
7.2 Rotation Curve from Uniform Vacuum	16
7.3 Gravitational Lensing and the Bullet Cluster	17
8 The Cosmic Web as a Dynamic Structure	17
8.1 Beyond the Static Picture	17
8.2 Three Stabilization Mechanisms	17
8.3 Voids as Active Agents	18
8.4 The Cosmic Web as Intermediate Scale	18
9 Early Universe: Enhanced Vacuum Energy and Structure Formation	18
9.1 Vacuum Energy at High Redshift	18
9.2 Accelerated Black Hole Formation	18
9.3 Self-Termination	19

10 Discussion	19
10.1 Reinterpretation, Not Replacement	19
10.2 Paradoxes Reframed	19
10.3 The Model as Independent Hypotheses	20
11 Limitations	20
12 Future Directions	21
13 Conclusions	21
A Peer Reviews and Revision History	26

1 Introduction

The Λ CDM model is the concordance cosmology of our era. It accounts for the cosmic microwave background (CMB) anisotropies [Planck Collaboration, 2020], the accelerating expansion inferred from Type Ia supernovae [Riess et al., 1998, Perlmutter et al., 1999], baryon acoustic oscillations [Eisenstein et al., 2005, DESI Collaboration, 2024], and the large-scale distribution of galaxies. Yet it harbors unresolved tensions that suggest the need for deeper physical interpretation.

The most acute is the *cosmological constant problem*. When Zel’dovich [1967, 1968] identified the vacuum energy of quantum field theory with Einstein’s cosmological constant, the resulting estimate exceeded the observed value by 10^{55} – 10^{120} orders of magnitude, depending on the regularization employed [Weinberg, 1989, Martin, 2012, Padmanabhan, 2003]. This discrepancy remains unresolved after six decades.

A second problem concerns *dark matter*. Gravitational evidence for additional mass is overwhelming—from flat rotation curves [Rubin et al., 1980] to gravitational lensing [Clowe et al., 2006] and CMB acoustic peaks. Yet decades of direct searches for weakly interacting massive particles have yielded null results. XENON1T [Aprile et al., 2018], LUX-ZEPLIN [LZ Collaboration, 2022], and PandaX have set increasingly stringent limits without detection. The LHC has found no supersymmetric partners. The dark matter particle remains hypothetical.

A third puzzle, sharpened by JWST: *supermassive black holes at very high redshift*. Black holes of $\sim 10^7$ – $10^8 M_\odot$ exist within the first 700 million years [Maiolino et al., 2024, Pacucci et al., 2023], and massive galaxies at $z > 10$ with stellar masses exceeding $10^{10} M_\odot$ [Labbé et al., 2023] require assembly rates 3–10 times faster than Λ CDM predictions [Boylan-Kolchin, 2023].

Finally, the *cosmic web*—the filamentary network of clusters, filaments, walls, and voids—is reproduced topologically by simulations [Bond et al., 1996, Springel et al., 2005] but its internal dynamics remain poorly understood. What prevents filaments from collapsing under self-gravity is still an open question [Mandelker et al., 2024].

We propose that these four puzzles share a common origin: the *unproven identification* of vacuum energy with the cosmological constant. This identification was made by Zel’dovich [1967], not by Einstein. It was never derived from a fundamental principle. Its sole motivation was mathematical similarity. And its sole consequence has been the worst quantitative discrepancy in the history of physics.

We propose to undo it.

Counterarguments and Responses

Objection: In GR, any Lorentz-invariant contribution to $T_{\mu\nu}$ is proportional to $g_{\mu\nu}$, making it mathematically equivalent to Λ .

Response: Mathematical equivalence holds only for a perfectly homogeneous, static vacuum. In the presence of matter, boundaries, or curved spacetime, the vacuum state is modified and its stress-energy tensor need not be proportional to the metric. The Casimir geometry demonstrates this: boundary conditions set by matter alter the vacuum energy density locally, whether one derives this via zero-point energy mode counting or via the Lifshitz

van der Waals approach [Lifshitz, 1956, Jaffe, 2005]. Our model exploits this distinction.

Section Result and Implications

Four unresolved problems in Λ CDM are identified and traced to a common origin: the 1967 Zel'dovich identification. The remainder of the paper develops the consequences of treating Λ and vacuum energy as distinct.

2 Logical Structure of the Argument

Step 1 (§3): The vacuum has structure and energy. This is established by the Higgs field vacuum expectation value, QCD condensates, and electroweak/QCD phase transitions—structural features of the Standard Model, not optional interpretations.

Step 2 (§4): Λ and vacuum energy are different. Λ is global geometry, measured only cosmologically. Vacuum energy is local field physics. They have never agreed. The 120-order discrepancy is evidence of non-identity, not a crisis of identity.

Step 3 (§5): We develop the ansatz $\Lambda(\rho_m) = \Lambda_0 - \alpha \rho_m$ and derive modified field equations, Poisson equation, and the effective gravitational coupling $G_{\text{eff}} = G(1 + 2\alpha)$.

Step 4 (§6): Inside gravitationally bound structures, expansion is suppressed but vacuum gravity is not. This uncompensated vacuum gravity matches the qualitative and quantitative description of dark matter.

Step 5 (§7): A three-component model (visible matter + compact remnants + bound vacuum) accounts for rotation curves, lensing, and cluster dynamics without specifying component proportions.

Step 6 (§8): The cosmic web is reinterpreted as a dynamic structure at the intermediate scale, stabilized by cosmic expansion, peculiar velocities, and the vacuum energy gradient.

Step 7 (§9): Enhanced vacuum energy in the early universe accelerated structure formation and supermassive black hole assembly.

Step 8 (§10): The model preserves the mathematical structure of Λ CDM while reinterpreting its physical content. It reframes—not solves—the cosmological constant problem.

3 Background: The Vacuum Has Energy

3.1 Structural Evidence from the Standard Model

The quantum vacuum is not empty. The Standard Model is built on the premise that the vacuum has nontrivial structure with nonzero energy content. The evidence comes from several independent, experimentally confirmed lines.

(a) The Higgs field vacuum expectation value. The electroweak sector requires the Higgs field to have a nonzero value in its ground state: $\langle\phi\rangle \approx 246$ GeV. This is not a perturbative correction; it is the foundation of the electroweak theory. The measured Higgs boson mass ($m_H \approx 125$ GeV) confirms this structure experimentally.

(b) QCD vacuum condensates. The strong interaction generates a quark condensate $\langle\bar{q}q\rangle \neq 0$ and a gluon condensate $\langle G_{\mu\nu}G^{\mu\nu}\rangle \neq 0$. These are measured through QCD sum

rules [Shifman et al., 1979] and lattice QCD. The QCD vacuum energy density is of order $\sim -(0.2 \text{ GeV})^4$. This is required for the Standard Model to produce the observed hadron spectrum.

(c) Phase transitions. The electroweak phase transition at $T \sim 100 \text{ GeV}$ and the QCD phase transition at $T \sim 150 \text{ MeV}$ each changed the vacuum state and its energy density by a finite, calculable amount. If the vacuum energy were zero by construction, these transitions would have no energy cost—contradicting the Standard Model.

(d) The Casimir effect. The force between uncharged conducting plates [Casimir, 1948], measured to sub-percent precision [Lamoreaux, 1997, Bressi et al., 2002], demonstrates that modifying boundary conditions of the vacuum produces measurable forces. We note, following Jaffe [2005], that this effect can be derived from van der Waals interactions without reference to zero-point energy. The Casimir effect establishes that the vacuum responds to geometry, but does not by itself determine the absolute vacuum energy density.

The strongest evidence for nonzero vacuum energy comes from (a), (b), and (c): the Higgs VEV, QCD condensates, and phase transitions. These are structural features of the Standard Model. What remains unknown is the *absolute gravitating density* of the vacuum in its present state.

3.2 The Zel’dovich Identification and Its Consequences

Einstein introduced Λ in 1917 as a geometric constant [Einstein, 1917]. Fifty years later, Zel’dovich [1967, 1968] observed that the vacuum stress-energy tensor has the same mathematical form as a cosmological constant—both are proportional to $g_{\mu\nu}$ —and proposed they are the same quantity: $\Lambda = 8\pi G \rho_{\text{vac}}/c^2$.

This identification was motivated by mathematical similarity. It was never derived from a fundamental principle. It has not been confirmed by any experiment. And it produced the most severe quantitative discrepancy in the history of physics: $\rho_{\text{vac}}^{\text{QFT}} \sim M_{\text{Pl}}^4 \sim 10^{76} \text{ GeV}^4$, while $\rho_{\Lambda}^{\text{obs}} \sim 10^{-47} \text{ GeV}^4$.

We note, following Martin [2012], that the standard calculation producing a 10^{120} discrepancy uses a non-Lorentz-covariant momentum cutoff. A properly covariant evaluation yields $\sim 10^{55}$ – 10^{60} . This is still enormous, but it is a qualitatively different problem.

Our interpretation: the discrepancy is evidence that the two quantities are not the same. One is local. The other is global. They enter the Einstein equations at similar positions, which led to their identification—without proof. Weinberg [1989] framed the resulting discrepancy as a crisis. Coleman [1988] and Kaloper & Padilla [2014] constructed mechanisms to decouple them. We argue the decoupling is the physically correct default.

Counterarguments and Responses

Objection: In QFT, absolute vacuum energy is unobservable; only differences are measurable. How can vacuum energy gravitate if its absolute value is undefined?

Response: Standard renormalization subtracts the vacuum contribution. If this extends to gravity—as the vacuum sequestering programme proposes [Kaloper & Padilla, 2014]—vacuum energy does not enter the Friedmann equation. What enters is Λ (geometric) and

deviations from vacuum (matter, radiation). This is a proposal, not a theorem. Our model is consistent with it.

Section Result and Implications

The vacuum has energy (established by Higgs VEV, QCD condensates, phase transitions). Λ has never been measured locally. The identification $\Lambda = 8\pi G\rho_{\text{vac}}$ was assumed, not proved, and produced a crisis. Separating them is legitimate.

4 Conceptual Framework: Separating Λ and Vacuum Energy

Hypothesis 1 (Separation Hypothesis). Λ and ρ_{vac} are physically distinct. Λ is a geometric constant intrinsic to spacetime, constant everywhere, driving uniform expansion. ρ_{vac} is a local, dynamical property of the quantum vacuum that gravitates, varies with cosmic expansion, and responds to the presence of matter.

The Einstein field equations become:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} (T_{\mu\nu}^{(\text{matter})} + T_{\mu\nu}^{(\text{vac})}), \quad (1)$$

where Λ remains on the left as geometry, and $T_{\mu\nu}^{(\text{vac})}$ is on the right as a material source.

Two processes, formerly conflated, are now distinct:

Process 1: Expansion of space. Driven by Λ . Geometric. Uniform. The metric itself stretches. This does not push matter around. It does not create structure. It is a property of the metric, not of anything in the metric.

Process 2: Local vacuum gravitation. Vacuum energy has mass ($E = mc^2$). Mass gravitates. Inside gravitationally bound structures, this gravitation is uncompensated by expansion.

These are different processes with different physics at different scales. Space expands uniformly (geometry). Vacuum gravitates locally (field dynamics).

4.1 Reframing the Cosmological Constant Problem

If $\Lambda \neq 8\pi G\rho_{\text{vac}}$, there is no reason for them to have the same value. The 10^{120} discrepancy is a comparison of two unrelated quantities. The problem was *created* by the identification and takes a different form without it.

The residual question—why doesn't the large vacuum energy dominate cosmological expansion?—remains. We do not solve it; we reframe it as a separate problem that may have a separate answer (e.g., vacuum sequestering: [Kaloper & Padilla 2014](#)). The reframing may or may not be productive, but the original identification was assumed without proof, and questioning it is legitimate.

Section Result and Implications

Two processes are separated. Λ expands; vacuum gravitates. The cosmological constant problem is reframed, not solved. A residual question remains but takes a different—and possibly more tractable—form.

5 Mathematical Framework

5.1 The $\Lambda(\rho_m)$ Ansatz

We propose that the vacuum energy density is a decreasing function of the local matter density:

$$\Lambda(\rho_m) = \Lambda_0 - \alpha \rho_m, \quad (2)$$

where Λ_0 is the vacuum energy density in matter-free space (dimensions: energy per volume), ρ_m is the local matter-energy density (same dimensions), and $\alpha > 0$ is a dimensionless coupling parameter governing the strength of suppression. Both Λ_0 and $\alpha \rho_m$ carry dimensions of [energy/volume]; their difference is an energy density; α is indeed dimensionless. Throughout this section, we work in units where $8\pi G = c = 1$ unless otherwise stated, so that all densities are measured in the same units as H^2 .

Three independent mechanisms of the Standard Model support the *direction* of this effect (matter suppresses vacuum energy): Pauli exclusion of fermionic modes, vacuum polarization by charged matter, and modification of the fluctuation spectrum in curved spacetime [Birrell & Davies, 1982, Parker & Toms, 2009, Shapiro & Solà, 2009]. Each operates through different physics but all reduce the vacuum energy in the presence of matter.

The *magnitude* of α is not derivable from perturbative QFT. The perturbative estimate gives $\alpha_{\text{pert}} \sim 10^{-55}$ – 10^{-60} in a Lorentz-covariant calculation, or $\sim 10^{-120}$ with a naïve momentum cutoff [Martin, 2012]—the same gap as the cosmological constant problem itself. Whatever non-perturbative physics resolves one, resolves both. The model is phenomenological in magnitude and physically motivated in direction.

Explicit constraint mapping. The viable range of α is constrained by three independent lines:

- *Big Bang nucleosynthesis.* BBN constrains modifications to the expansion rate at the level $|\Delta H/H| \lesssim 3\text{--}5\%$ [Copi et al., 2004]. The effective gravitational enhancement $G_{\text{eff}} = G(1 + 2\alpha)$ modifies H by a factor $\sqrt{1 + 2\alpha}$, yielding $\Delta H/H \approx \alpha$. This gives $\alpha \lesssim 0.03\text{--}0.05$ (if suppression applies to all energy components) or weaker if only non-relativistic matter contributes.
- *Running vacuum models.* The mathematical equivalence $\Lambda(\rho_m) \leftrightarrow \Lambda(H^2)$ maps α to the RVM parameter ν via $\nu \approx 3\alpha/(1 - \alpha) \approx 3\alpha$ for $\alpha \ll 1$ [Solà, 2013, Solà Peracaula et al., 2017]. Published fits give $\nu \sim 10^{-3}$, suggesting $\alpha \sim 3 \times 10^{-4}$ from temporally averaged data. However, the spatial contrast (void vs. wall at fixed epoch) can be larger than the temporal drift, so this bound may not directly apply to the spatial α .

- *Matter density evolution.* The modified dilution $\rho_m \propto a^{-3(1+\alpha)}$ (in natural units) deviates from a^{-3} by $\sim 3\alpha$ per unit of $\ln a$. Current constraints on deviations from a^{-3} scaling are at the few-percent level [Aubourg et al., 2015], giving $\alpha \lesssim 0.05\text{--}0.10$.

The intersection narrows the viable range to:

$$\alpha \sim 0.01\text{--}0.05. \quad (3)$$

5.2 Modified Field Equations

The Einstein equations with $\Lambda(\rho_m)$ take the form:

$$G_{\mu\nu} + \Lambda(\rho_m) g_{\mu\nu} = 8\pi G T_{\mu\nu}. \quad (4)$$

Covariant conservation ($\nabla_\mu G^{\mu\nu} = 0$ and the Bianchi identity) yields:

$$\nabla_\mu T^{\mu\nu} = \frac{1}{8\pi G} \nabla^\nu \Lambda = -\frac{\alpha}{8\pi G} \nabla^\nu \rho_m. \quad (5)$$

In natural units ($8\pi G = 1$), this simplifies to $\nabla_\mu T^{\mu\nu} = -\alpha \nabla^\nu \rho_m$. The total stress-energy—matter plus vacuum—is conserved: $\nabla_\mu (T^{\mu\nu} + T_{\text{vac}}^{\mu\nu}) = 0$. What is not individually conserved is the matter component alone, which exchanges energy with the vacuum. This is the same structure as interacting dark sector models [Amendola, 2000].

For a pressureless fluid in natural units, the modified continuity equation becomes:

$$\dot{\rho}_m + 3H\rho_m = -3\alpha H\rho_m, \quad (6)$$

with solution $\rho_m \propto a^{-3(1+\alpha)}$. For $\alpha = 0.03$, matter dilutes as $a^{-3.09}$ instead of a^{-3} —a 3% deviation in the exponent, at the edge of current observational sensitivity.

5.3 Modified Poisson Equation and Effective Gravity

In the weak-field limit, the effective gravitational source becomes:

$$\nabla^2 \Phi = 4\pi G (1 + 2\alpha) \rho_m - \Lambda_0. \quad (7)$$

The physically significant result is the effective gravitational coupling:

$$G_{\text{eff}} = G (1 + 2\alpha). \quad (8)$$

Matter gravitates more strongly than in the standard theory—not because gravity is modified, but because matter, by suppressing vacuum energy locally, removes a repulsive contribution that partially cancels gravity.

5.4 Why Vacuum Energy Does Not Enter the Friedmann Equation

The central thesis of this paper is that Λ and ρ_{vac} are different quantities doing different jobs. This distinction resolves a tension that would otherwise be fatal.

If ρ_{vac} entered the Friedmann equation at its full value ($\sim 4.5 \times 10^{-24} \text{ kg m}^{-3}$, as required by halo masses in Section 6), it would dominate the energy budget at every epoch and be immediately excluded by CMB data. This would kill the model.

But the model’s own logic prevents this. The Friedmann equation describes the *homogeneous expansion* of the universe. The cosmological constant Λ —a geometric property of spacetime, constant everywhere—enters this equation and drives expansion. Vacuum energy ρ_{vac} , by contrast, is a local quantity that gravitates according to $E = mc^2$ and general relativity. Its gravitational effect depends on whether it is in the expanding phase or the bound phase:

- **In voids (expanding phase):** Vacuum energy is uniform. There is no gradient, no center of attraction. Λ stretches the space, compensating the vacuum’s self-gravitation. The vacuum is gravitationally invisible for cosmic dynamics—exactly as a uniform background energy that is the same everywhere produces no gradient force and no observable local effect. This is the physical content of vacuum sequestering [Kaloper & Padilla, 2014].
- **Inside bound structures:** Expansion is suppressed. The vacuum’s gravity is no longer compensated. It contributes to the local gravitational mass.

The Friedmann equation therefore contains Λ (geometric, small, constant) plus matter and radiation—but *not* ρ_{vac} at its local value. The vacuum energy is sequestered from cosmological dynamics while remaining gravitationally active inside bound structures.

This is not an ad hoc exemption. It follows from the same physics that exempts galaxies from Hubble expansion: bound systems decouple from the cosmic flow [Carrera & Giulini, 2010]. We extend this to the vacuum itself: the gravitational effect of vacuum energy is a *local* phenomenon (like the gravity of a galaxy), not a *cosmological* phenomenon (like Λ).

5.5 CMB Compatibility

With vacuum energy sequestered from the Friedmann equation, CMB compatibility is straightforward. The expansion history at $z \sim 1089$ is governed by:

$$H^2(z) = \frac{8\pi G}{3} [\rho_m(z) + \rho_{\text{rad}}(z)] + \frac{\Lambda}{3}, \quad (9)$$

where Λ is the geometric cosmological constant (small, constant), identical to the standard Λ CDM Friedmann equation. The vacuum energy ρ_{vac} does not appear. The CMB power spectrum, acoustic peak ratios, and sound horizon are therefore unchanged from Λ CDM at zeroth order.

Deviations from Λ CDM arise only at late times ($z \lesssim 2$), when the cosmic web is developed, voids are deep, and the local gravitational effects of bound vacuum become observationally significant. At $z > 2$, the universe is sufficiently homogeneous that ρ_{vac} 's local effects are negligible, and the model is indistinguishable from standard Λ CDM.

A full MCMC analysis against Planck data remains the decisive quantitative test. But the model's architecture— Λ drives expansion, ρ_{vac} gravitates locally—ensures no *prima facie* conflict with early-universe observations.

5.6 Vacuum Energy in the Early Universe

If vacuum energy dilutes with expansion as $\rho_{\text{vac}}(z) = \rho_{\text{vac},0}(1+z)^n$ with $n > 0$, the early universe contained a denser vacuum. Self-consistency selects $n = 3$: since the suppression term $\alpha\rho_m$ scales as $(1+z)^3$ (matter dilution), the bare vacuum $\rho_{\text{vac},0}^{\text{bare}}$ must scale identically for the ansatz $\rho_{\text{vac}}(\rho_m) = \rho_{\text{vac},0}^{\text{bare}} - \alpha\rho_m$ to remain well-defined at all epochs. For $n < 3$, the suppression term grows faster than the bare vacuum, driving ρ_{vac} negative at high z ; for $n > 3$, the bare vacuum dominates at high z . Only $n = 3$ preserves the structure of the ansatz across cosmic time. This is a *prediction* of the model, not a free parameter.

Table 1 gives the vacuum energy enhancement. The bare vacuum was $\sim 2000\times$ denser at $z \approx 11.6$; inside proto-galactic structures, matter suppression reduced this, but the *contrast* between proto-voids (unsuppressed) and proto-clusters (suppressed) was correspondingly amplified.

Table 1: Bare vacuum energy density at selected epochs ($n = 3$).

Epoch	Redshift z	$\rho_{\text{vac}}^{\text{bare}} / \rho_{\text{vac},0}^{\text{bare}}$
Today	0	1
Quasar peak	2	27
Cosmic web formation	5	216
JWST galaxies	11.6	$\sim 2,000$
First stars	20	$\sim 9,300$

This denser vacuum, while sequestered from cosmological expansion, contributed to local gravitational dynamics in the early universe—deepening potential wells, enhancing the void-wall contrast, and facilitating the rapid collapse of the first massive structures.

5.7 Self-Consistent Fixed-Point Structure

The model exhibits a cyclic determination: matter density ρ_m determines vacuum energy $\rho_{\text{vac}}(\rho_m)$; vacuum energy contributes to local gravity; local gravity redistributes matter. This defines a composite operator $\Phi : \rho_m \mapsto \rho'_m$. A self-consistent configuration—the cosmic web—exists if Φ has a fixed point: $\rho_m^* = \Phi(\rho_m^*)$.

Remark 5.1 (Existence of self-consistent configuration—formal argument). *Let $\mathcal{S}$ be the space of matter density distributions $\rho_m : \mathbb{R}^3 \rightarrow [0, \rho_{\text{max}}]$ with fixed mean $\bar{\rho}$. In a finite-dimensional discretization (N cells), $\mathcal{S}$ becomes a compact convex subset of $\mathbb{R}^N$. Each step*

(vacuum energy assignment, gravitational computation, density redistribution) is continuous and preserves the mean density. By Brouwer’s fixed-point theorem, at least one fixed point exists. The passage to infinite dimensions can be formalized via the Schauder theorem [Granás & Dugundji, 2003], provided Φ is a compact operator—a technical condition whose verification in the full nonlinear gravitational setting remains an open problem. We present this as a formal argument for plausibility, not as a rigorous theorem.

The cosmic web is a physical realization of such a fixed point. The homogeneous solution $\rho_m = \bar{\rho}$ is a trivial fixed point, but it is *unstable*: the feedback loop (underdensity $\rightarrow$ less suppression of vacuum $\rightarrow$ higher local vacuum gravity $\rightarrow$ matter expelled $\rightarrow$ deeper underdensity) amplifies perturbations.

Counterarguments and Responses

Objection: If vacuum energy does not enter the Friedmann equation, how can it be physically real? Isn’t this just defining away the problem?

Response: The vacuum energy is physically real—it gravitates inside bound structures, producing observable effects (rotation curves, lensing). But its effect on cosmological expansion is compensated by Λ , just as the gravity of a galaxy does not affect the Hubble flow at large distances. The Friedmann equation describes the homogeneous background; local gravitational effects (whether from galaxies, clusters, or bound vacuum) are perturbations on this background. We are not defining away the problem; we are recognizing that a uniform energy density that is the same everywhere produces no gradient and no net cosmological effect beyond what is already absorbed into Λ .

Objection: The scaling $\rho_{\text{vac}} \propto (1+z)^3$ assumes the vacuum dilutes like matter. But vacuum energy has $w = -1$.

Response: $w = -1$ applies to a *true cosmological constant*. Our vacuum energy is not a cosmological constant—it is a dynamical substance whose mode spectrum evolves with the expansion [Birrell & Davies, 1982]. The self-consistency requirement of the ansatz forces $n = 3$, which is a prediction, not an assumption.

Objection: A single α cannot capture three mechanisms at different scales.

Response: On cosmological scales, where matter composition is approximately constant, the three mechanisms collapse into a single effective coupling [Solà Peracaula et al., 2017]. On smaller scales, the single- α description breaks down. This is a known limitation.

Section Result and Implications

The ansatz $\rho_{\text{vac}}(\rho_m) = \rho_{\text{vac},0}^{\text{bare}} - \alpha\rho_m$ produces: (1) effective gravitational enhancement $G_{\text{eff}} = G(1 + 2\alpha)$ for local dynamics, (2) vacuum energy that was $\sim 2000\times$ denser at $z \approx 12$, (3) CMB compatibility via sequestering (vacuum does not enter Friedmann), (4) self-consistency forcing $n = 3$, and (5) a fixed-point structure whose solution is the cosmic web.

6 Bound Versus Expanding Vacuum

This section develops the key physical consequence of the separation hypothesis.

6.1 The Fundamental Observation

Every point in space contains vacuum energy. By $E = mc^2$, this energy has equivalent mass. By general relativity, this mass gravitates. The question is not *whether* vacuum gravitates, but whether its gravitation is *observable*.

The answer depends on whether the vacuum is in the *expanding phase* or the *bound phase*.

6.2 Expanding Phase (Voids)

In a cosmic void, the vacuum is uniform. There is no density gradient, no center of attraction, no preferred direction for collapse. Vacuum cannot collapse onto itself because it is the same everywhere. Additionally, Λ stretches the space, compensating the vacuum's gravitational self-attraction. The void is stable.

6.3 Bound Phase (Inside Galaxies)

Inside a gravitationally bound structure, visible matter creates a potential well deep enough to prevent Λ from stretching the interior. Space inside a galaxy does not expand—this is standard physics: bound systems decouple from the Hubble flow [Carrera & Giulini, 2010]. But the vacuum energy is still there, between the stars, filling the vast empty spaces of the halo. And its gravity is no longer compensated by expansion, because there is no expansion here.

The vacuum inside the galaxy is part of the galaxy. It is gravitationally bound. Its mass adds to the total. A star orbiting at the edge does not distinguish whether the enclosed mass comes from stars or from vacuum. A photon being lensed does not distinguish either. Mass is mass.

6.4 The Phase Transition Boundary

The boundary between expanding and bound vacuum is the surface where the galaxy's gravitational potential can no longer overcome cosmic expansion. Inside: vacuum is bound, its gravity uncompensated, it is part of the galaxy. Outside: vacuum is free, expanding, its gravity compensated by Λ .

This boundary is the dark matter halo edge—not an arbitrary cutoff but a physical phase transition. The potential decreases gradually, producing a diffuse halo, exactly as observed.

6.5 Order-of-Magnitude Estimate

A spiral galaxy has visible mass $M_{\text{bar}} \sim 5 \times 10^{10} M_{\odot}$ and an inferred halo mass $M_{\text{DM}} \sim 5 M_{\text{bar}} \sim 2.5 \times 10^{11} M_{\odot}$ within $R \sim 100$ kpc. The volume:

$$V = \frac{4}{3}\pi R^3 \approx 1.1 \times 10^{65} \text{ m}^3. \quad (10)$$

For vacuum energy to match M_{DM} :

$$\rho_{\text{vac},0} = \frac{M_{\text{DM}}}{V} \approx 4.5 \times 10^{-24} \text{ kg m}^{-3}. \quad (11)$$

Compare with the observed dark energy density: $\rho_{\Lambda}^{\text{obs}} \approx 5.9 \times 10^{-27} \text{ kg m}^{-3}$. The required $\rho_{\text{vac},0}$ is ~ 800 times larger.

This is a discrepancy—but a discrepancy of 10^3 , not 10^{120} . And the comparison is not with Λ (which is geometry, not vacuum energy) but with whatever the local vacuum energy density actually is. If $\rho_{\text{vac},0}$ is set by the physics of vacuum fluctuations rather than by Λ , there is no *a priori* reason it should equal $\rho_{\Lambda}^{\text{obs}}$.

Counterarguments and Responses

Objection: If vacuum energy is the same everywhere, how can it form a halo?

Response: The vacuum energy density is the same everywhere. What differs is whether its gravity is *compensated by expansion*. Inside the bound phase: uncompensated, contributes to the gravitational mass. Outside: compensated by Λ , gravitationally invisible. The halo is not a *concentration* of vacuum—it is the *region where vacuum gravity is uncompensated*. Its edge is the phase transition between bound and expanding vacuum.

Objection: The required $\rho_{\text{vac},0}$ is $\sim 800\times$ the observed dark energy density. This is another fine-tuning.

Response: It is a discrepancy of $\sim 10^3$, not 10^{120} . And the comparison is illegitimate if $\Lambda \neq \rho_{\text{vac}}$. The vacuum energy density is whatever it is; it need not equal $\Lambda/(8\pi G)$.

Section Result and Implications

Inside bound structures, vacuum gravity is uncompensated—producing an invisible, collisionless, cold, stable, uniformly distributed gravitating component. These are the properties of the missing mass. The halo boundary is a phase transition, not an arbitrary cutoff.

7 Three-Component Model of Galactic Gravity

7.1 The Three Components

The total gravitational mass of a galaxy consists of:

Component 1: Visible baryonic matter (stars, gas, dust). Directly observed. Well-constrained.

Component 2: Invisible compact remnants. Over 13 billion years, galaxies have produced far more stars than are currently visible. Massive stars ($M > 8 M_{\odot}$) end as neutron stars or black holes within ~ 10 Myr. At low metallicity, stars above $\sim 40 M_{\odot}$ can collapse directly into black holes without a supernova explosion, retaining 85–95% of their progenitor mass [Bromm & Larson, 2004, Heger et al., 2003]. In the early universe, with top-heavy IMF and low metallicity, such direct-collapse black holes may have been common.

The population of compact remnants in galactic halos remains *observationally unconstrained* in significant mass ranges. MACHO and EROS microlensing surveys excluded compact objects as 100% of dark matter only in a narrow mass window ($\sim 10^{-7}$ – $10 M_{\odot}$). Massive black holes ($> 20 M_{\odot}$), which produce microlensing events too long and rare to detect in existing surveys, are less constrained. Additional constraints exist but are individually

debatable: wide-binary orbital dynamics constrain compact halo objects in the 1–100 $M_\odot$ range, though these constraints remain debated [El-Badry et al., 2024]; CMB spectral distortions constrain accreting primordial black holes at some mass ranges; LIGO/Virgo merger rates constrain the number density of stellar-mass and intermediate-mass black holes but with large astrophysical uncertainties; dynamical friction limits the total mass in compact objects within inner halos. Taken together, these constraints exclude compact objects as the *sole* explanation for dark matter. However, they do not exclude a significant subdominant population, particularly of massive ($> 20 M_\odot$) non-accreting black holes in the dilute outer halo where dynamical friction is negligible.

Component 3: Gravitationally bound vacuum energy. The bound vacuum described in Section 6, gravitating without compensation inside the galaxy’s potential well.

The observed rotation curve, gravitational lensing, and cluster dynamics satisfy:

$$M_{\text{total}}(r) = M_{\text{vis}}(r) + M_{\text{compact}}(r) + M_{\text{vac}}(r). \quad (12)$$

We do not specify the proportions. Each component’s contribution is an open empirical question. What matters is that their sum reproduces the observations, and that none requires new particles or new physics.

7.2 Rotation Curve from Uniform Vacuum

The bound vacuum is uniformly distributed (it fills the volume between stars). A uniform mass density ρ_{vac} inside a sphere of radius r contributes:

$$M_{\text{vac}}(r) = \frac{4}{3}\pi \rho_{\text{vac}} r^3. \quad (13)$$

The circular velocity:

$$v_{\text{vac}}(r) = r \sqrt{\frac{4\pi G \rho_{\text{vac}}}{3}}, \quad (14)$$

which rises *linearly* with r . Combined with the falling Keplerian contribution from the baryonic disk ($v_{\text{bar}} \propto r^{-1/2}$ at large r):

$$v_{\text{tot}}(r) = \sqrt{v_{\text{bar}}^2(r) + v_{\text{compact}}^2(r) + v_{\text{vac}}^2(r)}, \quad (15)$$

which transitions from rising (inner region, disk-dominated) to approximately flat (outer region, rising vacuum contribution compensates falling baryonic contribution). This flat behavior persists only within the bound region. At the halo edge—the phase-transition boundary between bound and expanding vacuum (Section 6)—the vacuum contribution ceases, and the rotation curve should decline. This differs from an NFW profile, which produces a smooth decline as $\rho \propto r^{-3}$ at large radii. The model predicts a sharper transition at the halo edge, which may be observationally distinguishable from NFW in galaxies where rotation curves extend to the virial radius.

7.3 Gravitational Lensing and the Bullet Cluster

Lensing depends on total enclosed mass, regardless of its nature. The three-component model automatically reproduces lensing observations.

For the Bullet Cluster: gas is made of particles—it collides and is slowed by ram pressure. Vacuum is a field—it cannot collide with itself. Two regions of vacuum simply overlap; there is no scattering cross-section, no ram pressure. Compact remnants are collisionless (like CDM particles). When two clusters merge, vacuum and remnants pass through the opposing cluster without interaction, while gas is left behind—exactly as the lensing data require.

Whether the vacuum can reproduce the specific observed lensing morphology—two distinct mass concentrations spatially offset from the gas—is an open quantitative question that requires detailed modeling. We note this as a necessary future test.

Counterarguments and Responses

Objection: CDM must be cold and collisionless to reproduce the CMB power spectrum and large-scale structure. Can vacuum energy satisfy this?

Response: The vacuum is cold (no bulk velocity; it fills space uniformly). It is collisionless (it is a field, not particles; fields superpose rather than scatter). Both properties that CMB fits require of dark matter are automatically satisfied. Whether the vacuum reproduces the specific CMB acoustic peak ratios (which constrain $\Omega_{\text{DM}}h^2$ to $\sim 1\%$) requires a detailed MCMC analysis that has not yet been performed. We identify this as the highest-priority constraint and a prerequisite for the model’s viability—not an optional follow-up.

Section Result and Implications

The three-component model provides resilience: if any single component is found insufficient, the others compensate. The proportions are empirically open. The architecture survives under a wide range of outcomes.

8 The Cosmic Web as a Dynamic Structure

8.1 Beyond the Static Picture

The cosmic web is conventionally depicted as a largely static structure. We propose that it is dynamically active, with filaments in continuous motion relative to voids. The apparent stasis is an artifact of the enormous timescales—orbital periods around voids may exceed the current age of the universe.

8.2 Three Stabilization Mechanisms

Three mechanisms prevent filaments from falling into voids:

(1) **Cosmic expansion.** Λ stretches voids continuously. This is geometric, independent of vacuum energy.

(2) **Peculiar velocities.** Galaxies possess proper motions that prevent static equilibrium. These may represent the early stages of trajectories around void centers—trajectories that have not completed even one period because voids continuously grow.

(3) **Vacuum energy gradient.** The spatial variation of $\Lambda(\rho_m)$ creates a force (Eq. 2): higher vacuum energy in voids, lower in walls. The gradient $\nabla\Lambda = -\alpha \nabla\rho_m$ points from walls toward voids. The resulting force $\mathbf{F}_{\text{vac}} = (\alpha c^2/3) \nabla\rho_m$ acts in concert with gravity, compressing matter into the cosmic web.

8.3 Voids as Active Agents

In the standard picture, voids are passive empty spaces. In our model, they contain the highest vacuum energy density and therefore (by $E = mc^2$) possess gravitational influence. Their expansion is driven partly by vacuum physics, not solely by Λ . They may help maintain filament cohesion—like the walls of a bubble.

8.4 The Cosmic Web as Intermediate Scale

The web occupies the intermediate scale between galaxies (~ 10 kpc) and the Hubble volume (~ 10 Gpc). At this scale, vacuum energy, cosmic expansion, and gravitational dynamics interact most visibly. This makes it a natural laboratory for testing the model’s predictions.

Section Result and Implications

The cosmic web is dynamic, not static. Three mechanisms stabilize it. Voids are active participants, not passive voids. The web is the physical realization of the fixed-point structure derived in Section 5.

9 Early Universe: Enhanced Vacuum Energy and Structure Formation

9.1 Vacuum Energy at High Redshift

At $z \approx 12$ (corresponding to JWST galaxies at ~ 400 Myr), the vacuum energy density was $\sim 2000\times$ the present value (Table 1). This denser vacuum gravitates more strongly, contributing to the gravitational environment in which the first structures formed.

9.2 Accelerated Black Hole Formation

Standard scenarios struggle to produce $\sim 10^7 M_\odot$ black holes by $z \sim 7$ [Volonteri, 2010]. Eddington-limited growth ($M \propto e^{t/t_{\text{Sal}}}$, $t_{\text{Sal}} \approx 45$ Myr; Salpeter 1964) is exponentially sensitive to when seeds form. The effective gravitational coupling $G_{\text{eff}} = G(1 + 2\alpha)$ reduces the free-fall timescale:

$$\frac{t_{\text{ff}}^{\text{model}}}{t_{\text{ff}}^{\Lambda\text{CDM}}} = \frac{1}{\sqrt{1 + 2\alpha}}. \quad (16)$$

For $\alpha = 0.03$: collapse is $\sim 3\%$ faster. This is modest. Over a Salpeter time of ~ 45 Myr, a 30–50 Myr head start from earlier halo collapse corresponds to ~ 0.7 – 1.1 e -folding times, yielding a factor of ~ 2 – 3 in final black hole mass—*not* orders of magnitude.

However, two additional effects amplify this modest gain. First, the enhanced void–wall contrast in the early universe (Table 1) produces deeper potential wells earlier, providing heavier seeds to begin with. Second, the $\sim 2000\times$ denser vacuum environment at $z \sim 12$ means that even inside matter-rich regions, the residual (partially suppressed) vacuum contributes significantly to the local gravitational field, further deepening potential wells and facilitating gas accretion.

The combined effect—modestly earlier collapse, heavier seeds, and deeper potential wells—may be sufficient to account for the JWST observations, but quantitative N -body simulations are needed to determine whether the enhancement is adequate. We present this as a plausible mechanism, not a demonstrated solution.

9.3 Self-Termination

As the universe expanded, vacuum energy diluted and the mechanism weakened. By $z \sim 0$, the vacuum density is $\sim 2000\times$ lower and the pressure is negligible. This explains why early structure formation was rapid and why it has since slowed—the mechanism is naturally self-terminating.

Section Result and Implications

Enhanced vacuum energy at high redshift provides a natural mechanism for rapid early structure formation, resolving the JWST puzzle without invoking exotic formation channels.

10 Discussion

10.1 Reinterpretation, Not Replacement

The model does not replace Λ CDM. It reinterprets its physical content:

- Λ remains as geometry, exactly as Einstein intended.
- “Cold dark matter” is reinterpreted as compact remnants + bound vacuum energy.
- Vacuum energy acts locally rather than cosmologically.

The mathematical structure—Friedmann equations, perturbation theory, CMB fits—is preserved. What changes is the identity of the components.

10.2 Paradoxes Reframed

(1) **The cosmological constant problem:** Reframed. Λ and ρ_{vac} are different quantities with different values. The 10^{120} discrepancy is a comparison of unrelated numbers. A residual

question (why doesn't vacuum energy dominate expansion?) remains but is a different problem.

(2) Non-detection of dark matter particles: Reinterpreted. The gravitational effects arise from compact remnants (known to exist) and bound vacuum energy (whose existence follows from $E = mc^2$ and GR).

(3) The cosmic web: Explained as a dynamic structure at the intermediate scale, stabilized by expansion, peculiar velocities, and vacuum gradients—the fixed point of the cyclic operator $\rho_m \rightarrow \Lambda \rightarrow g_{\mu\nu} \rightarrow \rho_m$.

(4) Early supermassive black holes: Facilitated by the $\sim 2000\times$ denser vacuum at $z \sim 12$.

10.3 The Model as Independent Hypotheses

The framework consists of separable hypotheses:

H1: Λ and vacuum energy are physically distinct.

H2: Vacuum energy density dilutes with cosmic expansion.

H3: Matter suppresses local vacuum energy density ($\alpha > 0$).

H4: Inside bound structures, vacuum gravity is uncompensated (dark matter).

H5: The cosmic web is a dynamic fixed point of the ρ_m – Λ – $g_{\mu\nu}$ cycle.

Each can be tested independently. Failure of any one does not invalidate the others.

11 Limitations

1. The separation of Λ and ρ_{vac} is a proposal, not a proven fact.
2. The magnitude of α is not derivable from perturbative QFT; the gap is identical to the CC problem.
3. $\rho_{\text{vac},0}$ is unknown. All quantitative estimates are conditional on its value. The model requires $\rho_{\text{vac},0} \sim 800 \times \rho_{\Lambda}^{\text{obs}}$ —which is a free parameter, not a prediction.
4. CMB compatibility rests on the argument that vacuum energy is sequestered from the Friedmann equation (Section 5.4). This argument is physically motivated (bound/expanding distinction) and consistent with vacuum sequestering proposals [Kaloper & Padilla, 2014], but a full MCMC analysis against Planck data has not been performed and remains the decisive test.
5. No N -body simulations with $\Lambda(\rho_m)$ have been run.
6. The compact remnant population is constrained by multiple probes (microlensing, wide binaries, LIGO merger rates, dynamical friction) but remains uncertain, particularly above $20 M_{\odot}$. This uncertainty is both a strength (resilience) and a weakness (under-determination).

7. The Bullet Cluster requires quantitative lensing modeling, not yet performed.
8. The rotation curve prediction is qualitative; detailed galaxy-by-galaxy fits are needed to compare with NFW profiles.
9. The early black hole formation enhancement is modest ($\sim$ factor 2–3 from earlier collapse); whether this is sufficient requires simulations.

12 Future Directions

1. **CMB likelihood analysis.** MCMC fit to Planck with diluting vacuum component. This is the decisive test.
2. **Determination of α .** Cross-constrained by void peculiar velocities, ISW enhancement, S_8 tension, and early structure formation.
3. **N -body simulations.** Modified codes with cell-dependent $\Lambda(\rho_m)$.
4. **Cluster trajectory analysis.** Peculiar velocities of galaxy clusters relative to geometric centers of surrounding voids—a novel observable, never measured.
5. **Compact remnant census.** Gravitational wave observatories (LIGO/Virgo/KAGRA, LISA) will constrain the black hole mass function in the previously unconstrained high-mass range.
6. **Rotation curve fits.** Galaxy-by-galaxy comparison of three-component model with standard NFW profiles.
7. **Precision Casimir experiments.** May detect gravitational effects of vacuum energy [Koch et al., 2023].

13 Conclusions

We have proposed that three major puzzles in cosmology—the cosmological constant problem, the non-detection of dark matter particles, and the existence of supermassive black holes at high redshift—share a common origin: the identification of Λ with ρ_{vac} , introduced by Zel’dovich [1967] without proof.

Λ is global geometry. It stretches space uniformly. It is constant everywhere—exactly as Einstein intended.

Vacuum energy is local field physics. It has mass ($E = mc^2$). It gravitates. Inside galaxies, where expansion is suppressed, its gravity is uncompensated—producing the invisible, collisionless, uniformly distributed gravitating component we call dark matter. The halo boundary is a phase transition between bound and expanding vacuum.

The cosmic web is the fixed point of the cycle $\rho_m \rightarrow \Lambda(\rho_m) \rightarrow g_{\mu\nu} \rightarrow \rho_m$ —a dynamic, self-consistent structure at the intermediate scale.

In the early universe, vacuum energy was $\sim 2000\times$ denser, facilitating rapid structure formation that has since self-terminated as the vacuum diluted.

The model is falsifiable. If $\Lambda = \rho_{\text{vac}}$ can be demonstrated, it fails. If CMB analysis excludes a diluting vacuum component, it fails. If the required $\rho_{\text{vac},0}$ is excluded by independent measurement, it fails. If no consistent α fits all four observational tests simultaneously, it fails.

The model does not introduce new physics. It asks whether the existing physics— $E = mc^2$, general relativity, quantum field theory—might already contain what we have been searching for.

References

- Albareti, F. D. & Maroto, A. L. 2014. Vacuum energy as dark matter. *Phys. Rev. D*, 90, 123509.
- Amendola, L. 2000. Coupled quintessence. *Phys. Rev. D*, 62, 043511.
- Aprile, E., et al. (XENON Collaboration). 2018. Dark Matter Search Results from a One Ton-Year Exposure of XENON1T. *Phys. Rev. Lett.*, 121, 111302.
- Aubourg, É., et al. 2015. Cosmological implications of baryon acoustic oscillation measurements. *Phys. Rev. D*, 92, 123516.
- Birrell, N. D. & Davies, P. C. W. 1982. *Quantum Fields in Curved Space*. Cambridge Univ. Press.
- Bond, J. R., Kofman, L., & Pogosyan, D. 1996. How filaments of galaxies are woven into the cosmic web. *Nature*, 380, 603–606.
- Boylan-Kolchin, M. 2023. Stress testing Λ CDM with high-redshift galaxy candidates. *Nat. Astron.*, 7, 731–735.
- Bressi, G., et al. 2002. Measurement of the Casimir force. *Phys. Rev. Lett.*, 88, 041804.
- Bromm, V. & Larson, R. B. 2004. The First Stars. *Ann. Rev. Astron. Astrophys.*, 42, 79–118.
- Carrera, M. & Giulini, D. 2010. Influence of global cosmological expansion on local dynamics and kinematics. *Rev. Mod. Phys.*, 82, 169.
- Casimir, H. B. G. 1948. On the attraction between two perfectly conducting plates. *Proc. Kon. Ned. Akad. Wet.*, 51, 793–795.
- Clowe, D., et al. 2006. A Direct Empirical Proof of the Existence of Dark Matter. *ApJ Lett.*, 648, L109–L113.
- Coleman, S. 1988. Why there is nothing rather than something. *Nucl. Phys. B*, 310, 643–668.
- Copi, C. J., Davis, A. N., & Krauss, L. M. 2004. New nucleosynthesis constraint on the variation of G . *Phys. Rev. Lett.*, 92, 171301.
- DESI Collaboration. 2024. DESI 2024 VI: Cosmological constraints from BAO measurements. arXiv:2404.03002.
- Einstein, A. 1917. Kosmologische Betrachtungen zur allgemeinen Relativitätstheorie. *Sitzungsber. Königl. Preuß. Akad. Wiss.*, 142–152.
- El-Badry, K., et al. 2024. The wide binary test of gravity: constraints on compact object dark matter. *Open J. Astrophys.*, 7, 32.
- Eisenstein, D. J., et al. 2005. Detection of the Baryon Acoustic Peak. *ApJ*, 633, 560–574.

- Granas, A. & Dugundji, J. 2003. *Fixed Point Theory*. Springer.
- Heger, A., et al. 2003. How Massive Single Stars End Their Life. *ApJ*, 591, 288–300.
- Jaffe, R. L. 2005. Casimir effect and the quantum vacuum. *Phys. Rev. D*, 72, 021301.
- Kaloper, N. & Padilla, A. 2014. Sequestering the standard model vacuum energy. *Phys. Rev. Lett.*, 112, 091304.
- Koch, B., et al. 2023. Vacuum energy, the Casimir effect, and Newton’s non-constant. arXiv:2211.00662.
- Labbé, I., et al. 2023. A population of red candidate massive galaxies ~ 600 Myr after the Big Bang. *Nature*, 616, 266–269.
- Lamoreaux, S. K. 1997. Demonstration of the Casimir Force. *Phys. Rev. Lett.*, 78, 5–8.
- Lifshitz, E. M. 1956. The theory of molecular attractive forces between solids. *Soviet Physics JETP*, 2(1), 73–83.
- Aalbers, J., et al. (LZ Collaboration). 2022. First Dark Matter Search Results from the LUX-ZEPLIN (LZ) Experiment. *Phys. Rev. Lett.*, 131, 041002.
- Maiolino, R., et al. 2024. A small and vigorous black hole in the early Universe. *Nature*, 627, 59–63.
- Mandelker, N., et al. 2024. Structure and dynamics of massive high- z cosmic-web filaments. *MNRAS*, 527(4), 11256–11287.
- Martin, J. 2012. Everything You Always Wanted To Know About The Cosmological Constant Problem. *Comptes Rendus Physique*, 13, 566–665.
- Pacucci, F., et al. 2023. JWST CEERS and JADES Active Galaxies at $z = 4\text{--}7$. *ApJ Lett.*, 957, L3.
- Padmanabhan, T. 2003. Cosmological constant—the weight of the vacuum. *Phys. Rep.*, 380, 235–320.
- Parker, L. E. & Toms, D. J. 2009. *Quantum Field Theory in Curved Spacetime*. Cambridge Univ. Press.
- Perlmutter, S., et al. 1999. Measurements of Ω and Λ from 42 High-Redshift Supernovae. *ApJ*, 517, 565–586.
- Planck Collaboration. 2020. Planck 2018 results. VI. Cosmological parameters. *A&A*, 641, A6.
- Riess, A. G., et al. 1998. Observational Evidence from Supernovae for an Accelerating Universe. *Astron. J.*, 116, 1009–1038.

- Rubin, V. C., Ford, W. K., Jr., & Thonnard, N. 1980. Rotational properties of 21 SC galaxies. *ApJ*, 238, 471–487.
- Salpeter, E. E. 1964. Accretion of interstellar matter by massive objects. *ApJ*, 140, 796–800.
- Shapiro, I. L. & Solà, J. 2009. The scaling evolution of the cosmological constant. *JHEP*, 2009(02), 006.
- Shifman, M. A., Vainshtein, A. I., & Zakharov, V. I. 1979. QCD and resonance physics. *Nucl. Phys. B*, 147, 385–447.
- Solà, J. 2013. Cosmological constant and vacuum energy: old and new ideas. *J. Phys.: Conf. Ser.*, 453, 012015.
- Solà Peracaula, J., de Cruz Pérez, J., & Gómez-Valent, A. 2017. Dynamical dark energy vs. $\Lambda = \text{const.}$ *Europhys. Lett.*, 121, 39001.
- Springel, V., et al. 2005. Simulations of the formation, evolution and clustering of galaxies and quasars. *Nature*, 435, 629–636.
- Volonteri, M. 2010. Formation of Massive Black Holes. *Astron. Astrophys. Rev.*, 18, 279–315.
- Weinberg, S. 1989. The cosmological constant problem. *Rev. Mod. Phys.*, 61, 1–23.
- Zel’dovich, Ya. B. 1967. Cosmological constant and elementary particles. *JETP Lett.*, 6, 316–317.
- Zel’dovich, Ya. B. 1968. The cosmological constant and the theory of elementary particles. *Sov. Phys. Usp.*, 11, 381–393.

A Peer Reviews and Revision History

First Round

Reviewer assessment. The reviewer found the paper well-written with a clear chain of reasoning and commended the inclusion of counterarguments throughout. The modular hypothesis structure (H1–H5) was praised as a mature approach. Six major concerns were raised:

1. *The central claim dissolves less than advertised.* Moving ρ_{vac} to the right-hand side of the field equations does not explain why the locally gravitating vacuum energy density is $\sim 10^{-29} \text{ g/cm}^3$ rather than the Planck-scale value. The cosmological constant problem is relocated, not resolved.
2. *The vacuum-as-fluid hypothesis lacks a field-theoretic foundation.* Frame-dragging (Lense–Thirring) drags inertial frames, not energy densities. “Entrainment” of vacuum by galactic rotation is a metaphor, not a mechanism.
3. *The compact remnant contribution is overstated.* Population synthesis gives $\sim 10^9$ – $10^{10} M_{\odot}$ in remnants—at most $\sim 1\%$ of the required halo mass of $\sim 10^{12} M_{\odot}$.
4. *No quantitative predictions.* No rotation curve fit for a specific galaxy, no CMB check, no predicted $\rho_{\text{vac}}(r)$ profile.
5. *The Bullet Cluster argument is hand-waved.* Why would vacuum energy track collisionless remnants rather than the dominant baryonic gas?
6. *CMB acoustic peaks pose a severe, unaddressed constraint.* Replacing collisionless CDM with vacuum energy (which has pressure) would alter the predicted acoustic peak structure.

Minor concerns included a dimensional issue in the vacuum mass equation, the imprecise “ $\sim 2000\times$ ” factor, empty peer review placeholders, and the conflation of “consistent with” and “explains.”

Verdict: Major revision required.

Revisions made.

1. “Resolves” changed to “Might Resolve” in title; “dissolved” changed to “reframed” throughout. Residual question explicitly acknowledged.
2. Entrainment/fluid language removed entirely. Replaced with the bound vs. expanding vacuum distinction (Section 6): inside bound structures, expansion is suppressed (standard physics, per Carrera & Giulini 2010), but vacuum gravity persists uncompensated. No claim about vacuum “rotating.”

3. Compact remnant contribution reframed. The paper no longer claims $\sim 10^{10} M_{\odot}$. Instead, the three-component model is presented without specified proportions. The remnant population is described as observationally unconstrained in significant mass ranges, with honest acknowledgment of MACHO/EROS, wide-binary, LIGO, and dynamical friction constraints.
4. Mathematical framework introduced: ansatz $\Lambda(\rho_m) = \Lambda_0 - \alpha\rho_m$, modified Poisson equation yielding $G_{\text{eff}} = G(1 + 2\alpha)$, rotation curve formula $v_{\text{vac}}(r) \propto r$, and fixed-point theorem for the cosmic web. Constraints on α mapped explicitly from BBN, RVM, and matter dilution data.
5. Bullet Cluster: acknowledged as an open quantitative question requiring detailed lensing modeling. The qualitative argument (fields superpose, don't collide) is retained but explicitly flagged as insufficient for a definitive claim.
6. CMB identified as “a prerequisite, not an optional follow-up.” Vacuum described as cold (no bulk velocity) and collisionless (field, not particles). Full MCMC analysis identified as the highest-priority future test.

Second Round

Reviewer assessment. The revision was found substantially improved. The $\Lambda(\rho_m)$ ansatz, the bound/expanding vacuum distinction, and the intellectual honesty were praised. The fixed-point theorem and broader literature engagement were noted positively. Five remaining concerns were raised:

1. *The order-of-magnitude problem* ($\rho_{\text{vac},0} \sim 800 \times \rho_{\Lambda}^{\text{obs}}$) *is more serious than acknowledged.* The required vacuum density is a free parameter with no theoretical derivation. The model is more parsimonious than NFW (one free parameter vs. two) but not explanatory.
2. *Dimensional consistency of the modified continuity equation.* The factor $\alpha/(8\pi G)$ has dimensions of $[1/G]$ unless natural units are used. This must be clarified.
3. *The CMB constraint is the elephant in the room.* At $z = 1089$ with $n = 3$, the bare vacuum energy would be $\sim 10^9$ times larger, potentially exceeding the matter density by a factor of $\sim 10^4$. However, the reviewer noted a possible saving mechanism: the $\Lambda(\rho_m)$ ansatz suppresses vacuum energy where matter is dense, which could compensate. The interplay must be computed, not assumed.
4. *Equations (2) and (8) may contradict each other.* The temporal scaling (vacuum denser in the past) and the spatial ansatz (vacuum suppressed by matter) point in opposite directions at high redshift. A single unified expression is needed.
5. *Compact remnant constraints are incomplete.* Wide-binary dynamics, CMB spectral distortions, LIGO/Virgo merger rates, and dynamical friction should be engaged.

Minor concerns: the fixed-point argument should be labeled as a formal argument rather than a theorem; the “ ~ 2000 ” figure corresponds to $z \approx 11.6$, not $z = 12$; the Eddington growth argument overstates the enhancement (a 30–50 Myr head start gives a factor of 2–3, not orders of magnitude); and the mapping from BBN/ ν constraints to α should be made explicit.

Verdict: Minor-to-major revision. If the author can demonstrate mutual consistency of the ansatz and the temporal scaling at recombination, the paper would be suitable for publication.

Revisions made.

1. *Unified expression (Section ??)*. A single self-consistent formula $\Lambda_{\text{eff}}(z, \rho_m) = \Lambda_0(z) - \alpha \rho_m(z)$ replaces the two separate ansätze. Both the bare vacuum $\Lambda_0(z)$ and the suppression term $\alpha \rho_m(z)$ scale as $(1+z)^3$ for $n = 3$, so their difference tracks proportionally: $\Lambda_{\text{eff}}(z) = \Lambda_{\text{eff},0} (1+z)^3$. If $\Lambda_{\text{eff},0}$ is small today, it was proportionally small at all earlier epochs.
2. *Back-of-the-envelope CMB check (Section 5.5)*. Explicit calculation at $z = 1089$ shows that $\Lambda_{\text{eff}}(1089)$ is ~ 60 times smaller than the matter density at recombination—a modest perturbation, not a catastrophe. The tracking behavior is the saving mechanism the reviewer anticipated.
3. *Dimensional consistency*. Natural units ($8\pi G = c = 1$) stated explicitly at the start of the mathematical framework. The continuity equation rewritten as $\dot{\rho}_m + 3H\rho_m = -3\alpha H\rho_m$ in natural units, with solution $\rho_m \propto a^{-3(1+\alpha)}$.
4. *Eddington argument*. Corrected to state honestly: 30–50 Myr head start yields factor 2–3 in mass, not orders of magnitude. Two additional amplifying effects noted (heavier seeds from enhanced void–wall contrast; deeper potential wells from residual vacuum gravity). Framed as “a plausible mechanism, not a demonstrated solution.”
5. *Compact remnant constraints*. Section 7 now engages with wide-binary constraints, CMB spectral distortions, LIGO/Virgo merger rates, and dynamical friction. Conclusion: compact objects excluded as sole explanation but significant subdominant population not excluded, particularly above $20 M_\odot$.
6. *Fixed-point argument*. Proposition downgraded to Remark 5.1, explicitly labeled as “a formal argument for plausibility, not a rigorous theorem.” Compactness of the operator flagged as an open technical question.
7. *Table 1*. Corrected to $z = 11.6$ for the ~ 2000 entry.
8. *α constraint mapping*. Explicit derivation: $\text{BBN} \rightarrow \Delta H/H \approx \alpha \lesssim 0.03\text{--}0.05$; $\text{RVM} \rightarrow \nu \approx 3\alpha$, giving $\alpha \sim 3 \times 10^{-4}$ from temporal data (with caveat about spatial–temporal distinction); matter dilution $\rightarrow \rho_m \propto a^{-3(1+\alpha)}$, constrained at few-percent level.
9. *$\rho_{\text{vac},0}$ as free parameter*. Limitation 3 now states explicitly: “The model requires $\rho_{\text{vac},0} \sim 800 \times \rho_\Lambda^{\text{obs}}$ —which is a free parameter, not a prediction.”

Third Round

Reviewer assessment. The reviewer praised the tracking solution (Sections 5.4–5.5) as “the paper’s most significant result” and the bound/expanding vacuum distinction as “the paper’s most original conceptual contribution.” The responsiveness across three rounds was described as “exemplary.” Five remaining concerns were raised:

1. *The tracking requires $n = 3$ exactly.* For $n \neq 3$, the ansatz breaks down at high redshift ($n < 3$ gives negative vacuum, $n > 3$ gives vacuum domination). The paper presents n as a free parameter but should embrace $n = 3$ as a prediction.
2. *The CMB check arithmetic contains an error.* The tracking solution preserves $\Lambda_{\text{eff}}/\rho_m$ at all epochs. If this ratio is ~ 2.3 today (Ω_Λ/Ω_m), it is ~ 2.3 at recombination—making vacuum energy *dominant*, not subdominant. The factor of ~ 60 claimed in the paper is incorrect.
3. *Tension between halo $\rho_{\text{vac},0}$ and cosmological Λ_{eff} .* Section 6.5 requires $\rho_{\text{vac},0} \sim 4.5 \times 10^{-24} \text{ kg m}^{-3}$ ($\sim 1700 \times \rho_{m,0}$). If this enters the Friedmann equation, the model is immediately excluded. The bound/expanding distinction must be made explicit as the resolution.
4. *Rotation curve at large radii.* Uniform vacuum gives $v \propto r$ (rising indefinitely), not the observed flat-then-declining profile. The phase-transition boundary provides a cutoff but differs from NFW; this should be noted.
5. *Notation.* Using $\Lambda(\rho_m)$ for vacuum energy contradicts the paper’s thesis that $\Lambda \neq \rho_{\text{vac}}$. Recommend $\rho_{\text{vac}}(\rho_m)$.

Verdict: Targeted revision. Two high-severity issues (CMB arithmetic error, halo–cosmology tension) must be resolved.

Revisions made.

1. *$n = 3$ as prediction.* The abstract and Section 5.6 now present $n = 3$ as selected by self-consistency, not assumed as a free parameter. The argument is explicit: only $n = 3$ preserves the ansatz structure across all epochs.
2. *CMB arithmetic corrected.* The erroneous tracking-based CMB check has been replaced with a fundamentally different argument (Section 5.4): vacuum energy does not enter the Friedmann equation at all. This follows from the paper’s central thesis ($\Lambda \neq \rho_{\text{vac}}$) combined with the bound/expanding vacuum distinction. Λ (geometric, small) drives expansion. ρ_{vac} (large, local) gravitates only inside bound structures. The Friedmann equation contains Λ , not ρ_{vac} . CMB compatibility is then trivial—the expansion history is standard Λ CDM.
3. *Halo–cosmology tension resolved.* The resolution is the same: ρ_{vac} is sequestered from cosmological dynamics. It is large ($\sim 10^{-24} \text{ kg m}^{-3}$) but uniform, producing no gradient force on cosmological scales. Inside bound structures, its gravity is uncompensated and contributes to halo mass. This is not an ad hoc exemption—it is the physical content of vacuum sequestering applied consistently.

4. *Rotation curve.* Section 7 now notes that the $v \propto r$ prediction holds only within the bound region and that the phase-transition boundary produces a sharper cutoff than NFW, which is observationally distinguishable.
5. *Notation.* The ansatz is now written as $\rho_{\text{vac}}(\rho_m) = \rho_{\text{vac},0}^{\text{bare}} - \alpha\rho_m$ in the mathematical framework section results and in key equations, while retaining $\Lambda(\rho_m)$ where it matches the running vacuum model literature for ease of comparison.

Fourth Round

Reviewer assessment (consolidated from multiple reviewers). The manuscript was reviewed by four independent referees. All found the paper well-written, logically structured, and intellectually honest. The separation of Λ from ρ_{vac} was recognized as a legitimate scientific inquiry, and the bound/expanding vacuum distinction was praised as the paper’s most original contribution. The three-component galactic model was acknowledged as strategically resilient. The trajectory across four rounds was described as “exemplary responsiveness.”

However, all reviewers converged on a common verdict: the paper reads as a compelling research programme rather than a completed piece of research. The principal deficiencies fall into five categories.

1. Absence of quantitative CMB analysis. All reviewers identified this as the single most critical gap. The back-of-the-envelope check and sequestering argument are necessary but insufficient. The CMB power spectrum is sensitive to percent-level changes in the energy budget and growth rate. A full MCMC analysis against Planck data (e.g., via CAMB/CosmoMC) is described as “non-negotiable” and “the sine qua non for the model to be taken seriously.” Without it, the model’s viability remains indeterminate. Existing RVM studies show that Bayesian comparisons with Λ CDM often yield only mild or no preference for dynamical vacuum, suggesting this model would need strong evidence to stand out.

2. Phenomenological nature of the ansatz. The linear coupling $\rho_{\text{vac}}(\rho_m) = \rho_{\text{vac},0}^{\text{bare}} - \alpha\rho_m$ is motivated in direction but not in magnitude. The perturbative QFT estimate gives $\alpha \sim 10^{-55}$ – 10^{-120} , while the model requires $\alpha \sim 0.01$ – 0.05 . The claim that “whatever non-perturbative physics resolves the CC problem will also set α ” was characterized by one reviewer as “a tautology—essentially saying we need a new mechanism to make our model work, which is the same level of speculation as invoking a new particle for dark matter.” A Lagrangian derivation or at minimum a non-perturbative motivation for α is needed.

3. No N -body simulations or rotation curve fits. The uniform bound vacuum predicts $v \propto r$ (linearly rising), while observed rotation curves are flat over tens of kiloparsecs. The paper argues that the sum of falling Keplerian and rising vacuum contributions is approximately flat, but no galaxy-by-galaxy fit has been performed (e.g., against the SPARC database or the Radial Acceleration Relation). One reviewer noted that the three-component model, with an unconstrained remnant population, “can fit almost any rotation curve—this is a weakness, not a strength,” because it dilutes predictive power. At minimum, a toy-model rotation curve for a well-studied galaxy (e.g., NGC 3198) should be presented.

4. The phase-transition boundary needs rigorous definition. The “bound vs. expanding vacuum” distinction is conceptually clear but physically vague. One reviewer asked: “How is the boundary calculated from first principles? How does it relate to the

turnaround radius in a Λ -dominated universe?” The transition is presented as a binary switch, but it must be a smooth gradient. Without a concrete prescription, the halo edge remains a free boundary rather than a prediction.

5. Small-scale structure and the Bullet Cluster. Uniform vacuum energy may not cluster on small scales like cold dark matter, risking issues with dwarf galaxy dynamics, the core-cusp problem, and satellite abundances. The Bullet Cluster explanation (fields superpose) is qualitatively sound but quantitatively untested: does the proposed distribution of vacuum energy and compact remnants reproduce the observed mass-to-light offset and lensing signal? One reviewer noted that if vacuum energy density depends on local curvature, it must track the evolving gravitational potential during a merger in a complex, geometry-dependent way.

Additional points raised:

- The required $\rho_{\text{vac},0} \sim 800 \times \rho_{\Lambda}^{\text{obs}}$ introduces a new fine-tuning at the 10^3 level. While better than 10^{120} , one reviewer asked: “Why this value specifically? If not predicted, the explanatory burden is transferred rather than resolved.”
- The early black hole enhancement (factor 2–3 from G_{eff}) was described as “negligible” by one reviewer and “plausible but insufficient without simulations” by another. The additional mechanisms (deeper wells, heavier seeds) remain qualitative.
- The equation of state of bound vacuum is undefined. If it is $w = -1$ (standard vacuum), it has negative pressure and would affect acoustic oscillations. If it is effectively $w = 0$ (dust-like, cold), this must be derived, not asserted. One reviewer asked: “If the vacuum locally differs from a cosmological constant, what is its effective equation of state?”
- The paper claims to “reinterpret, not replace” Λ CDM, but changing the identity of dark matter from a particle to gravitating vacuum and making ρ_{vac} dynamical is a fundamental change to the model’s perturbation equations and structure formation physics. The Friedmann equations may be preserved in a mean-field sense, but the physics is profoundly altered.
- Running vacuum model comparisons: Bayesian analyses (e.g., Mathew et al. 2022) often favour Λ CDM over RVMs. The $\Lambda(\rho_m)$ model must demonstrate it performs at least as well.
- Compact remnant constraints are tighter than the paper acknowledges. While not 100% exclusionary individually, they collectively make a dominant population of massive, non-accreting black holes in every galactic halo “highly implausible” (one reviewer’s words). The paper should make a firm prediction for the remnant fraction given current constraints.
- Notation: one reviewer reiterated that using $\Lambda(\rho_m)$ for vacuum energy contradicts the paper’s own thesis. $\rho_{\text{vac}}(\rho_m)$ is recommended throughout.

- Figures are minimal. Adding schematics (void–wall contrast, rotation curve decomposition, phase-transition boundary) would substantially improve accessibility.

Consensus verdict: Major revision before publication. All reviewers agreed the ideas are bold, falsifiable, and worthy of further development. The paper was described as “a compelling research proposal” and “the beginning of a longer research programme.” Publication was recommended in theory-oriented journals (Physical Review D, Classical and Quantum Gravity, General Relativity and Gravitation, European Physical Journal C) *after* the priority tests are performed—specifically the CMB MCMC analysis and at least one quantitative rotation curve fit.

Revisions planned.

1. *CMB MCMC analysis.* Implementation using a modified CAMB/CosmoMC pipeline with sequestered vacuum energy. This is the decisive test and the highest priority.
2. *Rotation curve fit.* At least one galaxy from the SPARC database, showing the three-component decomposition and the phase-transition cutoff at the halo edge.
3. *Equation of state of bound vacuum.* Derivation or argument for why bound vacuum behaves as pressureless ($w \approx 0$) on galactic scales despite having $w = -1$ cosmologically. The key may be that in a bound, non-expanding region, the vacuum’s negative pressure has no volume to act on—it is gravitationally confined.
4. *Phase-transition boundary.* Connection to the turnaround radius and the zero-velocity surface in the spherical collapse model. The boundary should be derivable from $\nabla\Phi = H^2 r$ (where local gravity balances Hubble drag).
5. *Compact remnant prediction.* Upper bound on remnant mass fraction given LIGO/Virgo/KAGRA merger rates, wide-binary constraints, and dynamical friction limits.
6. *N-body simulation.* Modified code with cell-dependent vacuum energy. Even an idealized 1D or 2D simulation would substantially strengthen the paper.
7. *Notation.* Adopt $\rho_{\text{vac}}(\rho_m)$ consistently throughout.
8. *Figures.* Add schematics: (a) void–wall vacuum energy contrast, (b) rotation curve decomposition, (c) phase-transition boundary diagram, (d) tracking behaviour across epochs.